Forecasting the closing stock price of Reliance Industries Using Time Series Analysis

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Project objectives

The objective of our project is to forecast the closing stock prices of Reliance Industries using time series modelling techniques such as exponential smoothing and ARIMA

What is Time series?

A time series is a sequence of data points that occur in successive order over some period of time.

What’s the need of time series forecasting for stocks?

Time series forecasting is important because it helps us predict future stock prices based on historical data, enabling better investment decisions.

Provides insights into future market trends, volatility, and risk management.

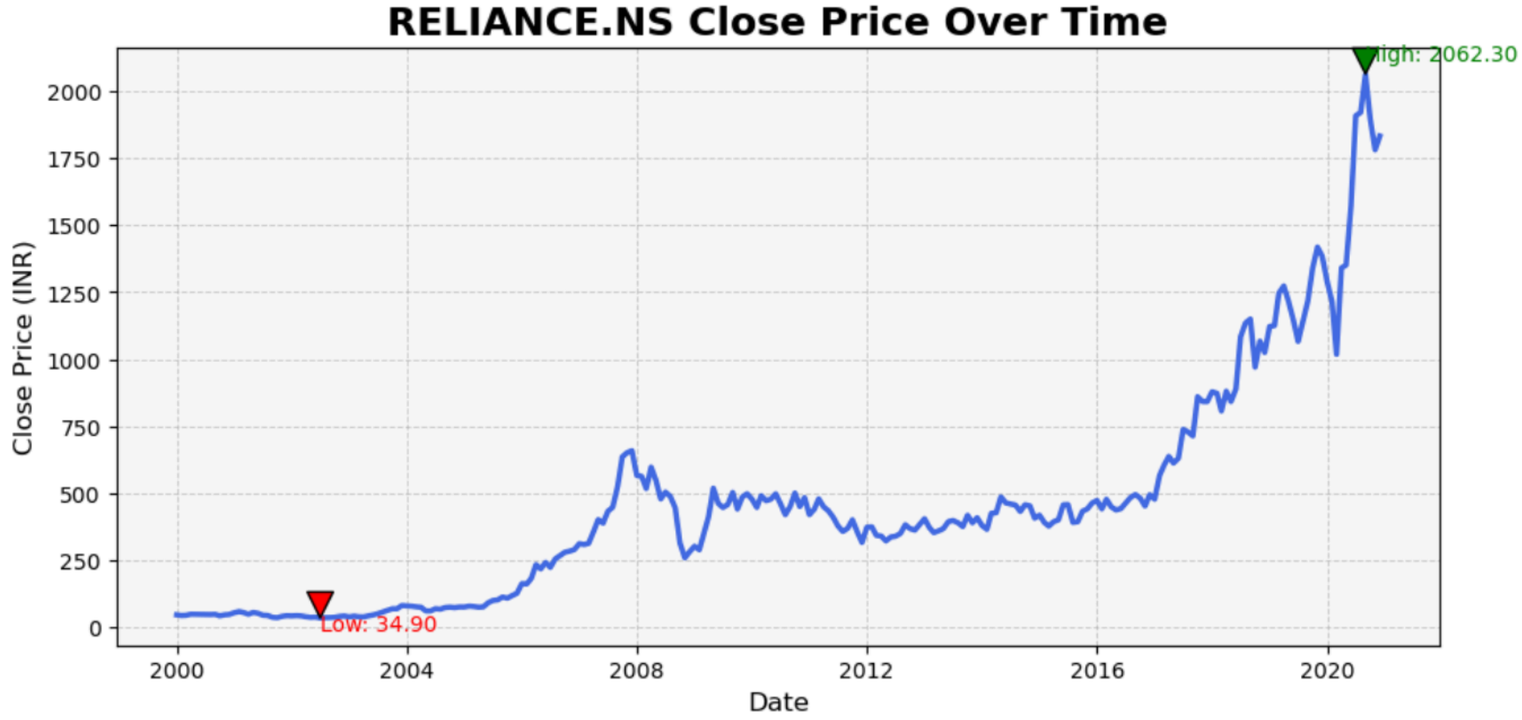
Apart from stocks, time series analysis is also used in various other domains such as business, finance, healthcare, weather predictions and more.



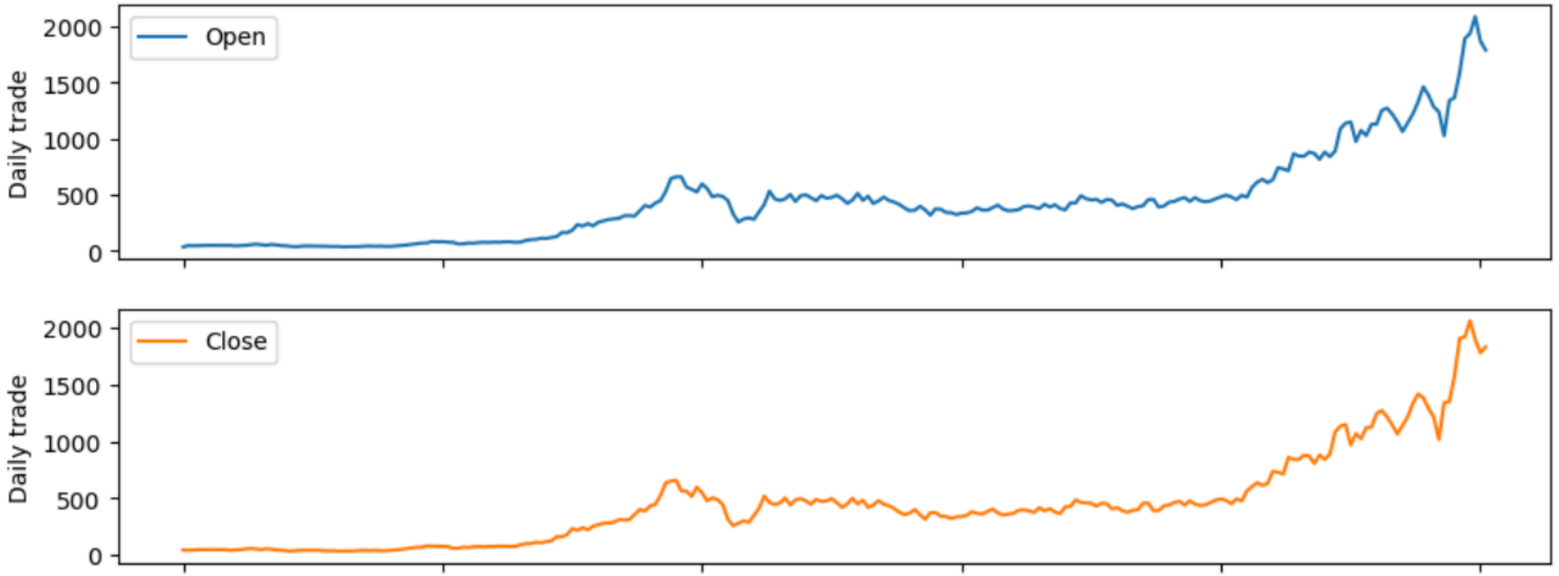
About Data

The dataset consists of Monthly Reliance stock prices from 2000 till 2023

	Date	Close
0	2000-01-01	46.863815
1	2000-02-01	44.393166
2	2000-03-01	45.192913
3	2000-04-01	49.277344
4	2000-05-01	49.113110

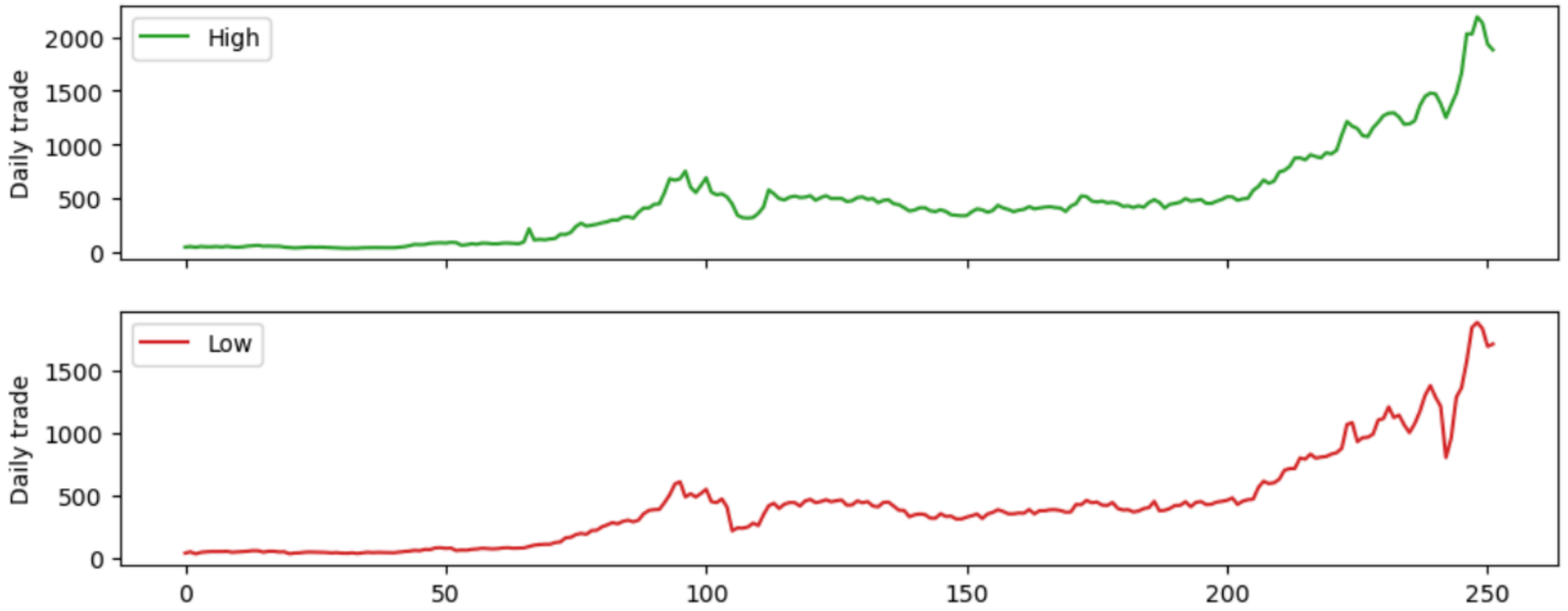


About Data



Opening and Closing Prices

High and Low Prices



Data Preprocessing

```
#Number of rows and columns in the dataset (monthly data)
data.shape
```

```
(252, 7)
```

The span of the dataset is 252 rows spread over 7 columns

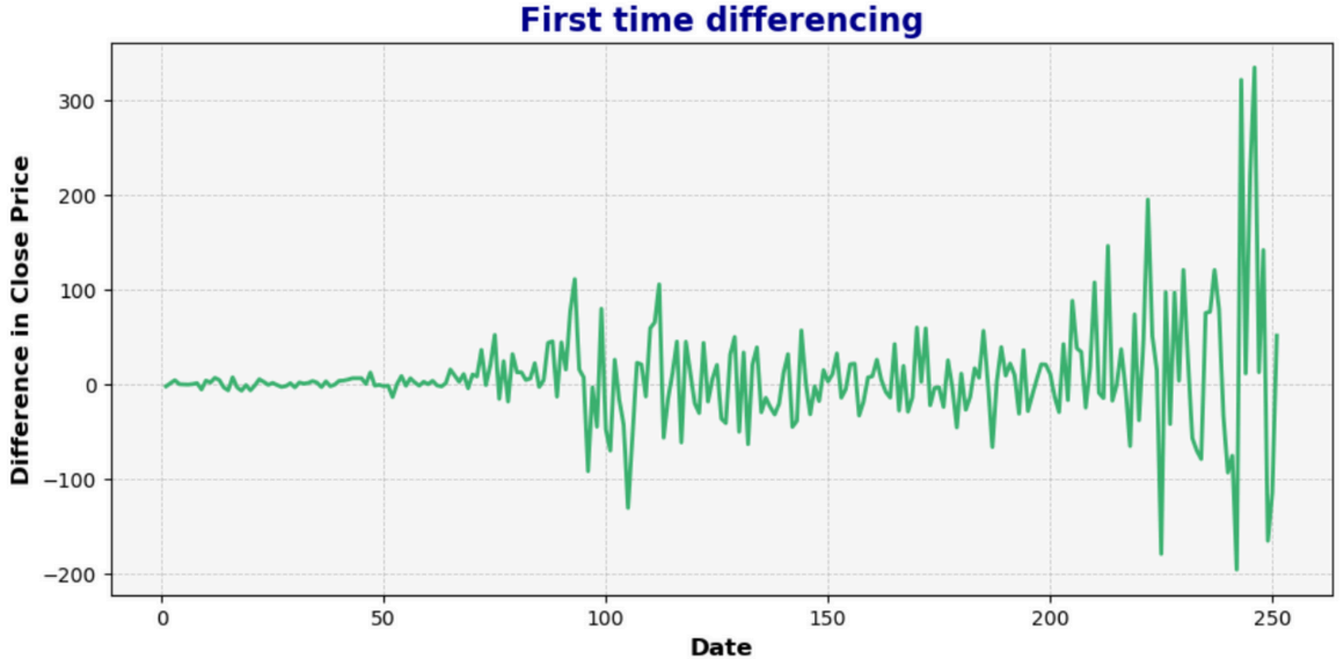
```
data.isnull().sum()
```

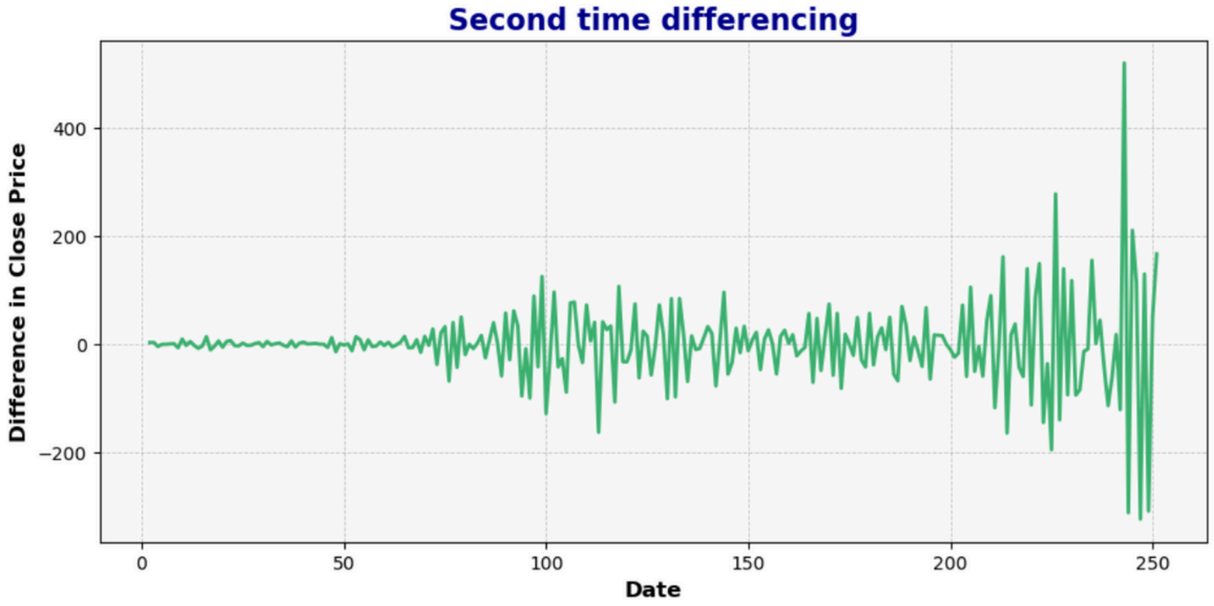
```
Date         0
Open         0
High         0
Low          0
Close        0
Adj Close    0
Volume       0
dtype: int64
```

There are no null values in the dataset so there is no need for imputation by mean or median

Making Data Stationary

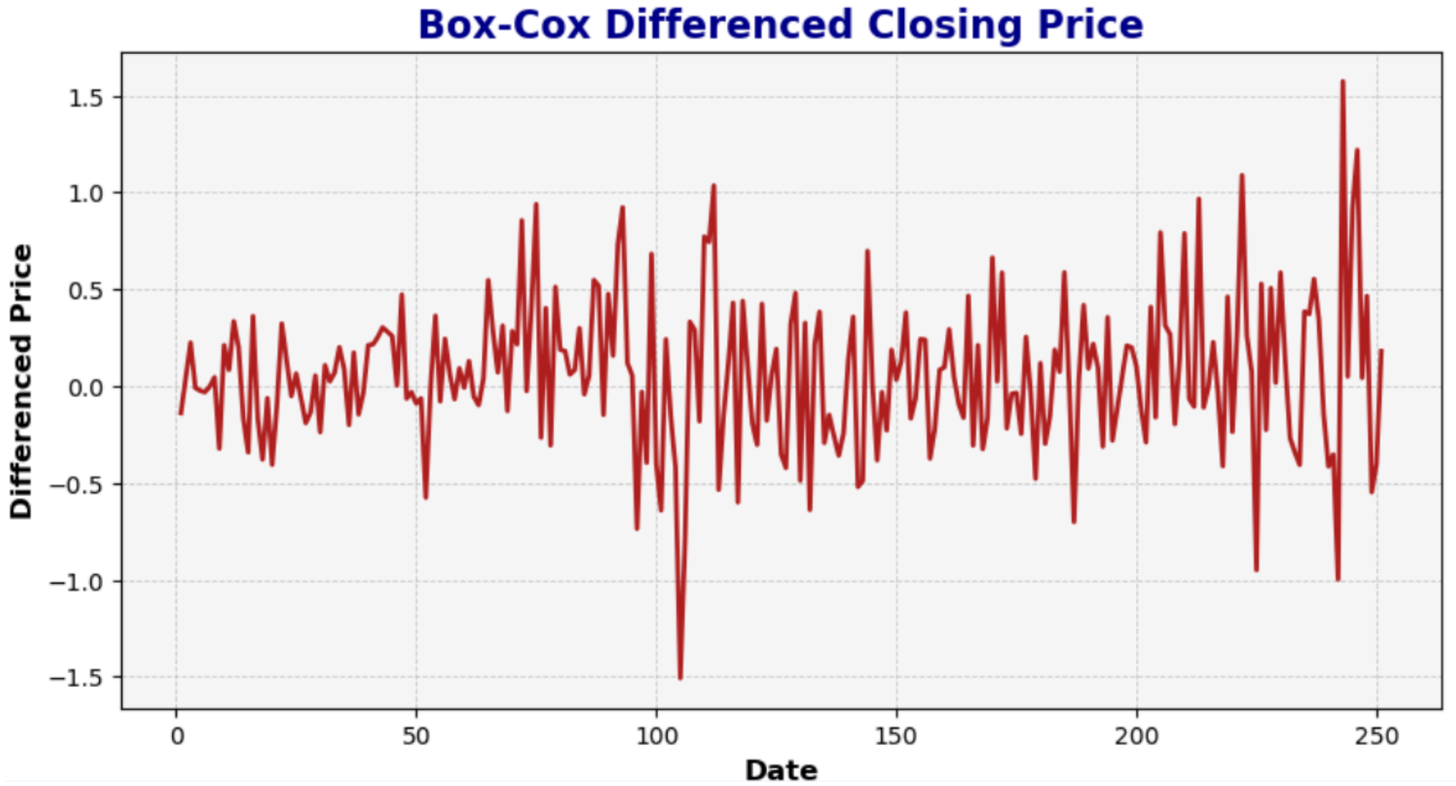
As it is evident from the previous time series plot of the closing prices it can be seen that the data is non-stationary so we proceed with differencing.





Since even after differencing it can be observed that the data does not show a constant mean and variance indicating that it is non-stationary hence box cox test is applyed which helps in transforming data to stabilize variance, making it more suitable for statistical modeling.

Making Data Stationary

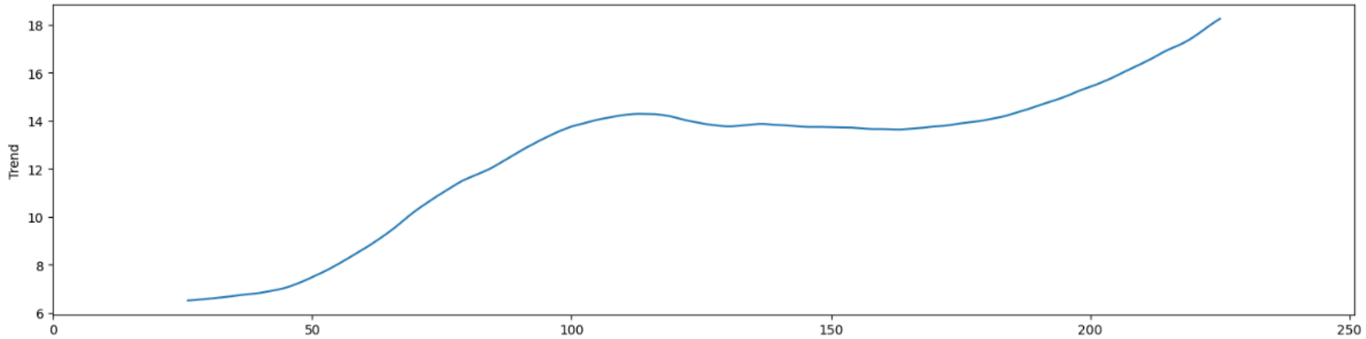



```
ADF Test Statistic : -15.385486307563845
p-value : 3.363496976226922e-28
#Lags Used : 0
Number of Observations Used : 250
strong evidence against the null hypothesis(Ho), reject the null hypothesis. Data has no unit root and is stationary
```

The data is stationary after applying differencing on the box cox transformed data which is confirmed by the ADF test

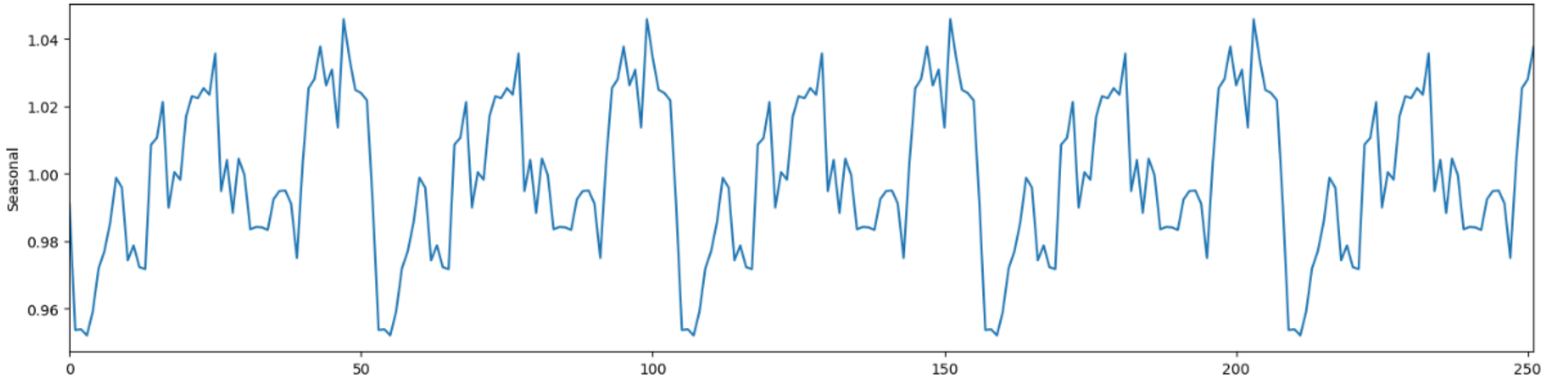
Seasonal Decomposition

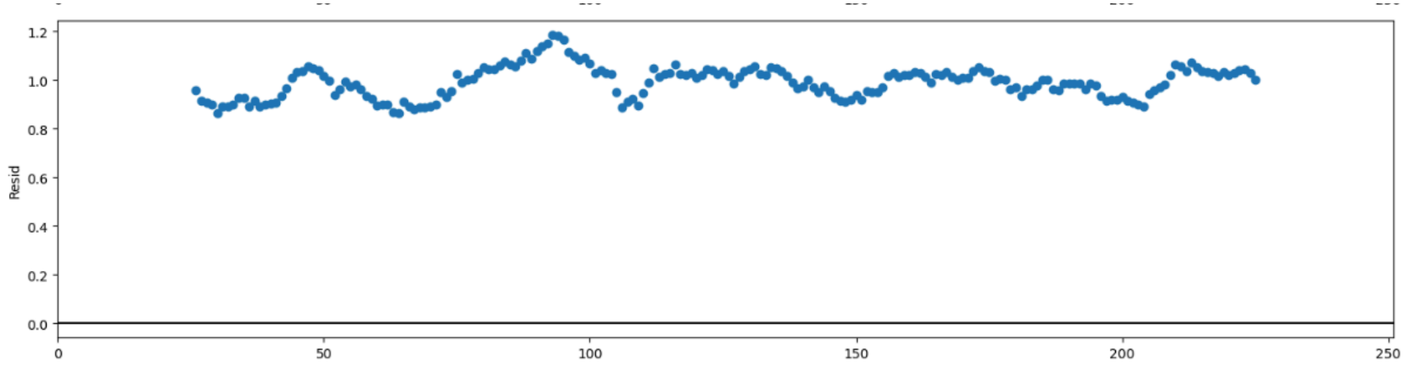
Multiplicative Decomposition



- Overall long-term movement in the time series data
- Captures the upward or downward trend over time
- Smoothing out the short-term fluctuations.

- Repeating pattern or seasonality in the data
- Cyclic behavior that repeats over a specific period

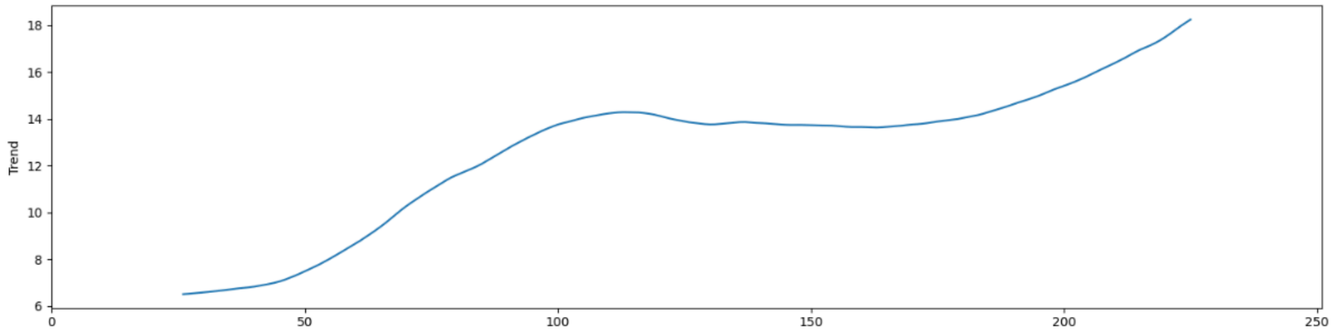




- Represents the residuals or the random noise after removing the trend and seasonal components from the original data
- Whatever is left is just often representing irregular or random fluctuations.

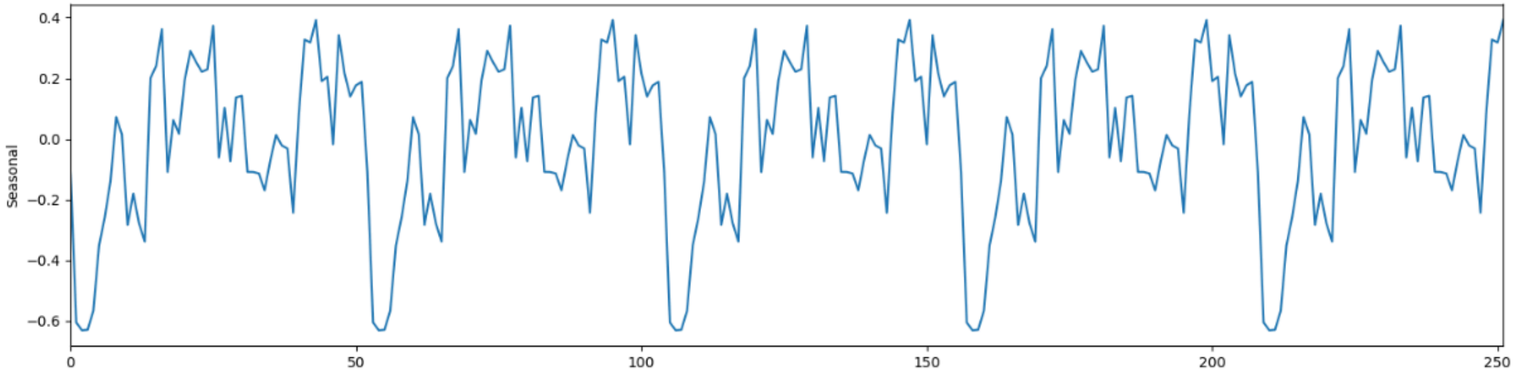
Seasonal Decomposition

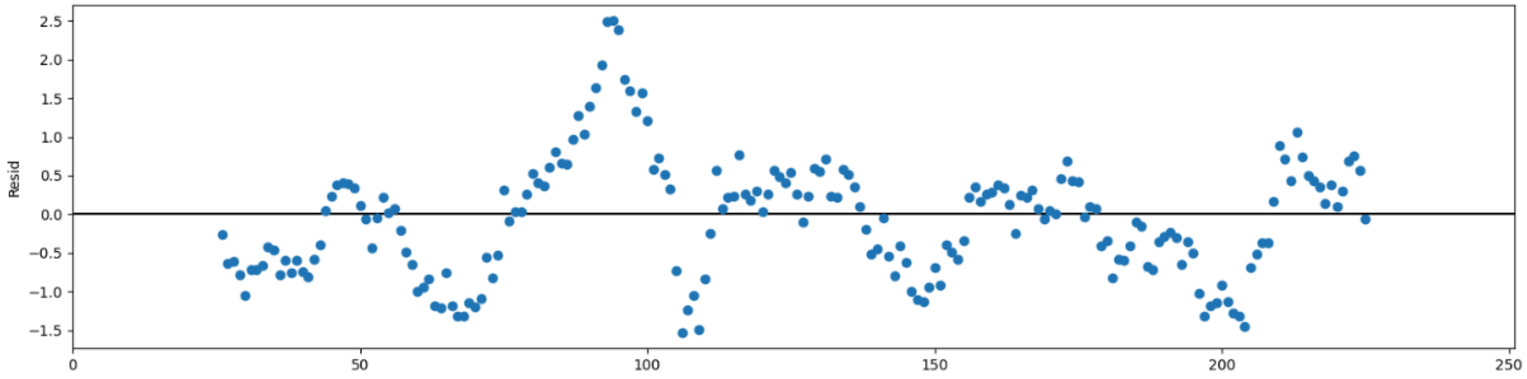
Additive Decomposition



- An upward trend indicates that the variable is increasing over time

- The seasonal pattern shows regular fluctuations, possibly due to periodic events
- These fluctuations are relatively consistent, indicating predictable changes at regular intervals.





- There is visible autocorrelation
- This suggests that the time series has components that are not well captured by the current decomposition model.
- The residuals in the plot show some structure rather than behaving like random noise.

Holt Winters Exponential Smoothing

Holt-Winters Exponential Smoothing, also known as Triple Exponential Smoothing, is a time series forecasting method that extends simple exponential smoothing by adding components to handle trends and seasonality. It's particularly useful for data with both a trend and a seasonal pattern.

1. The exponentially smoothed series, or level estimate:

$$L_t = \alpha \frac{Y_t}{S_{t-s}} + (1 - \alpha)(L_{t-1} + T_{t-1})$$

2. The trend estimate:

$$T_t = \beta(L_t - L_{t-1}) + (1 - \beta)T_{t-1}$$

3. The seasonality estimate:

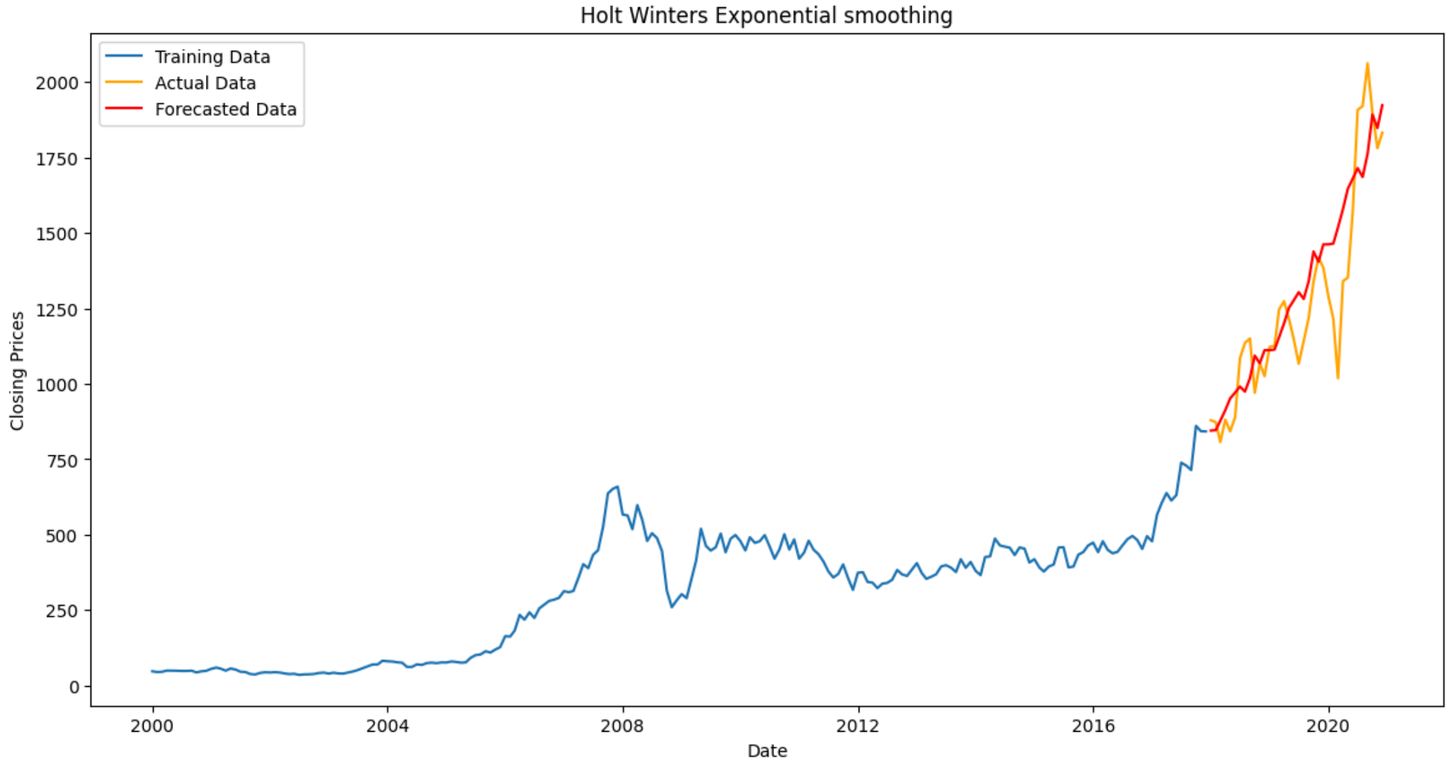
$$S_t = \gamma \frac{Y_t}{L_t} + (1 - \gamma)S_{t-s}$$

4. The forecast for p periods into the future:

$$\hat{Y}_{t+p} = (L_t + pT_t)S_{t-s+p}$$

Holt Winters Exponential Smoothing

- The training data exhibits a clear upward trend over time, with periods of volatility. The Holt-Winters model, which accounts for trends and seasonality, seems to capture this trend and the short-term fluctuations well.
- The close alignment between the actual data and the forecasted data indicates that the Holt-Winters model is effectively predicting future values. It successfully follows the upward trend and the periodic fluctuations.



Root Mean Square Error: 339.0897935571036
MAPE: 17.09%

The Equations:

$$\ell_t = 0.9152518 \times \frac{y_t}{s_{t-m}} + (1 - 0.9152518)(\ell_{t-1} + b_{t-1})$$

$$b_t = 0.0001207(\ell_t - \ell_{t-1}) + (1 - 0.0001207)b_{t-1}$$

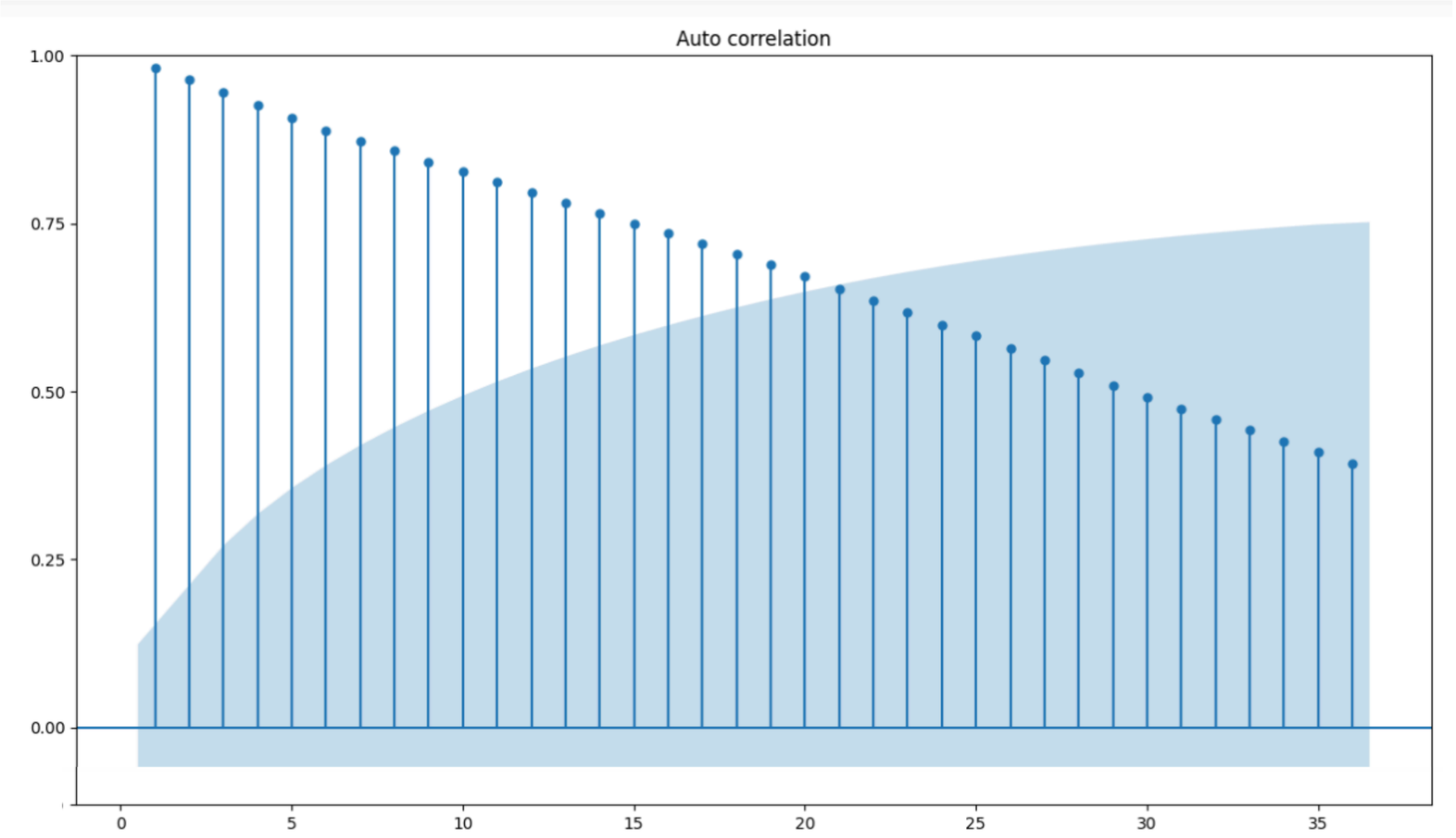
$$s_t = 0.0836115 \times \frac{y_t}{\ell_t} + (1 - 0.0836115)s_{t-m}$$

The forecasted Equation:

$$\hat{y}_{t+h|t} = (\ell_t + h \times 1.0106525) \times s_{t+h-m(k+1)}$$

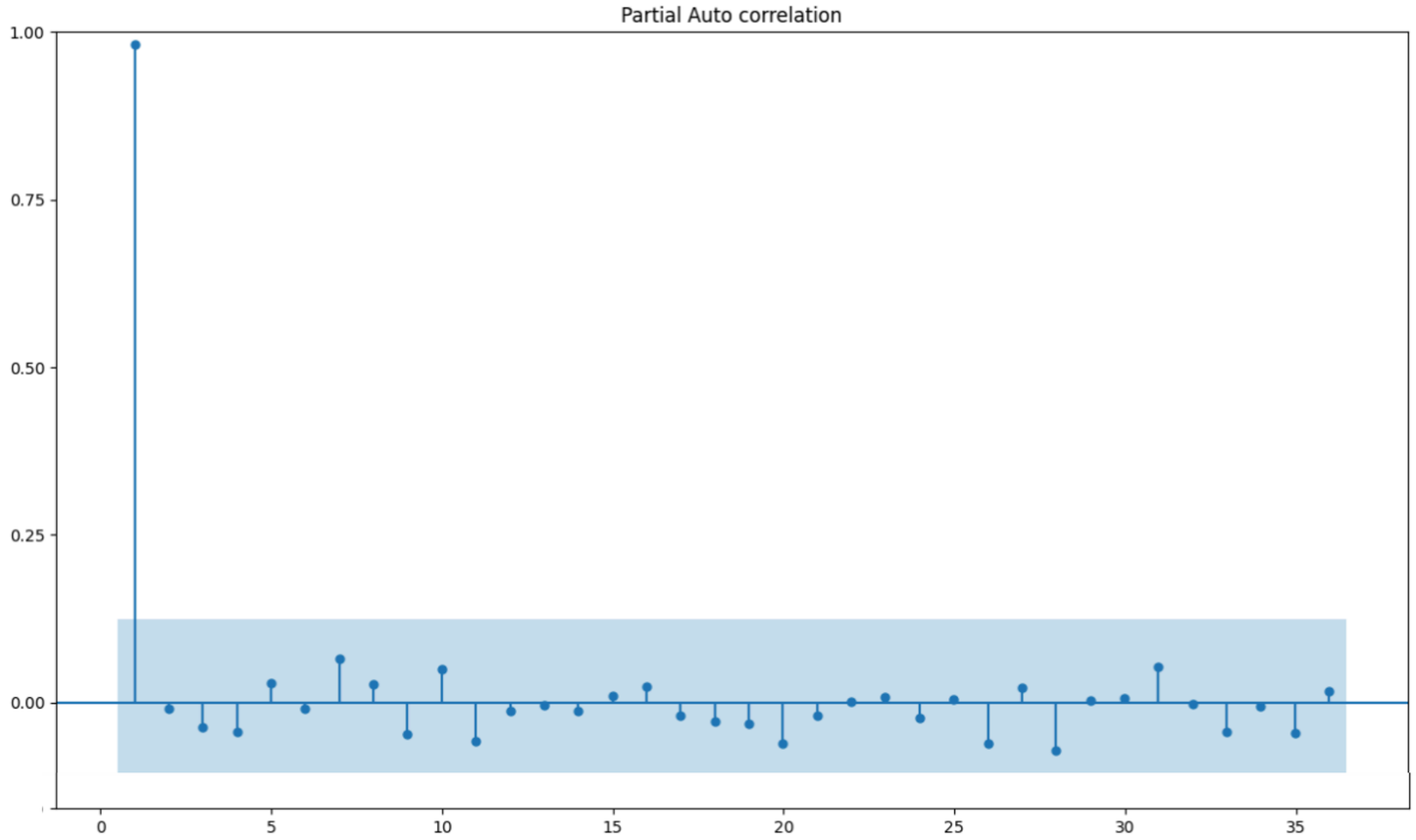
ACF Graph

- The ACF graph shows the autocorrelation of the time series at different lags.
- The graph depicts a slow decay.
- This pattern is common in time series with a trend or seasonal component.



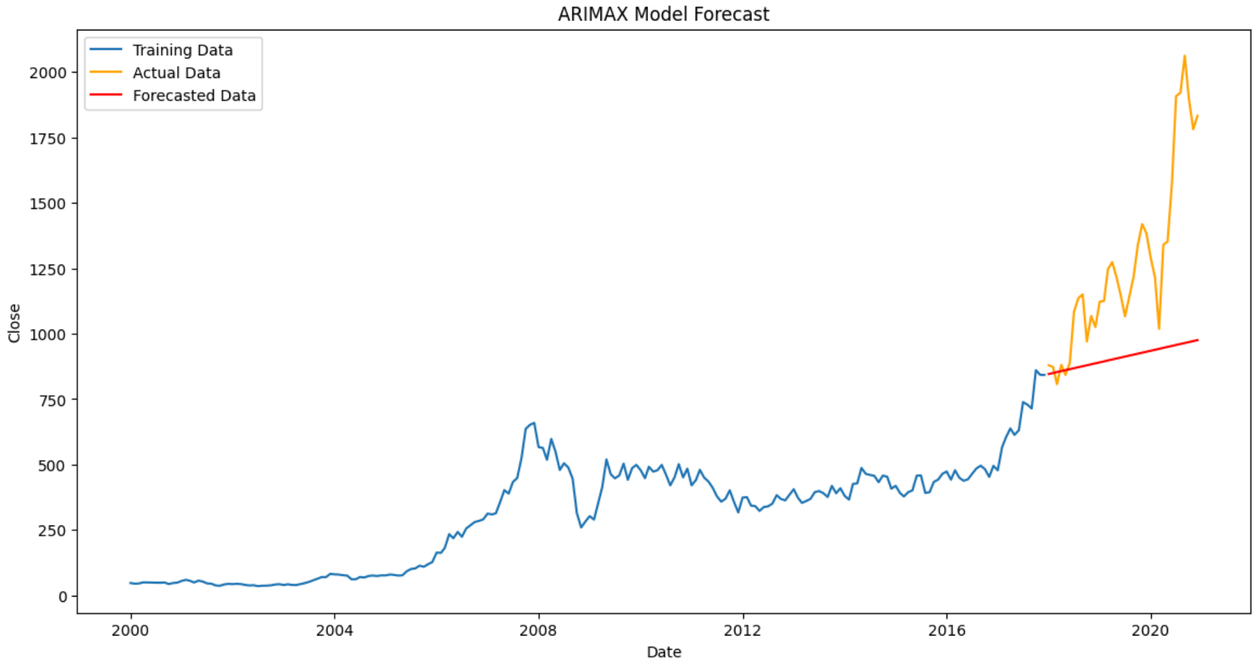
PACF Graph

- The PACF graph shows the partial autocorrelation of the time series at different lags, which measures the correlation of the residuals after removing the effects of earlier lags.
- In the PACF graph, the significant spike at lag 1, followed by a rapid drop to near zero, suggests that an AR(1) model might be appropriate.



ARIMA

- The forecasted values seem to follow an upward trend similar to the actual observed values , but with a noticeable underestimation in the recent data points.



The Equation:

$$y_t = 3.6990 + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q} + \epsilon_t$$

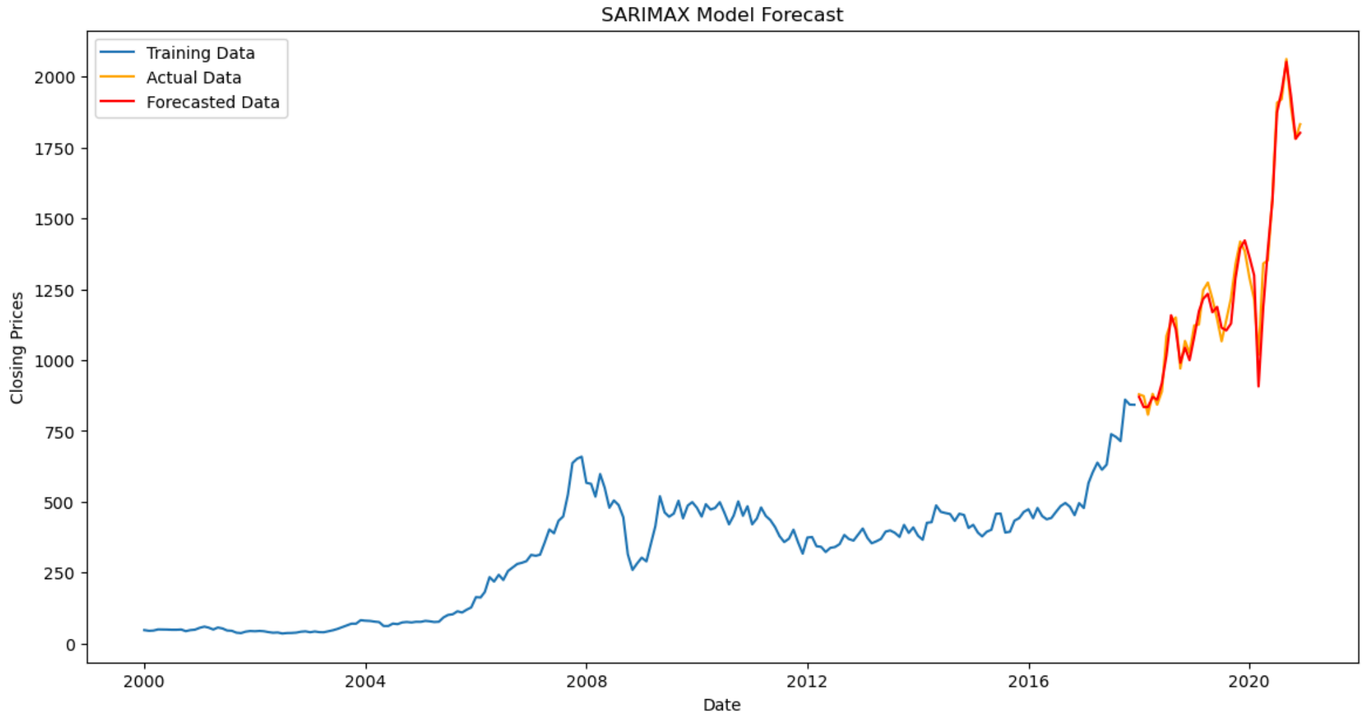
Root Mean Squared Error: 464.1834555470885
MAPE: 24.43%

After testing with various other ARIMA models, The best model is

```
ARIMA(0,1,0)(0,0,0)[0] intercept   : AIC=2110.026, Time=0.03 sec
```

SARIMA

- **The training data exhibits a clear upward trend over time, with some periods of volatility and fluctuations. This trend continues into the actual data, which the model captures well in its forecast.**



Root Mean Square Error: 50.896177
MAPE : 3.43%

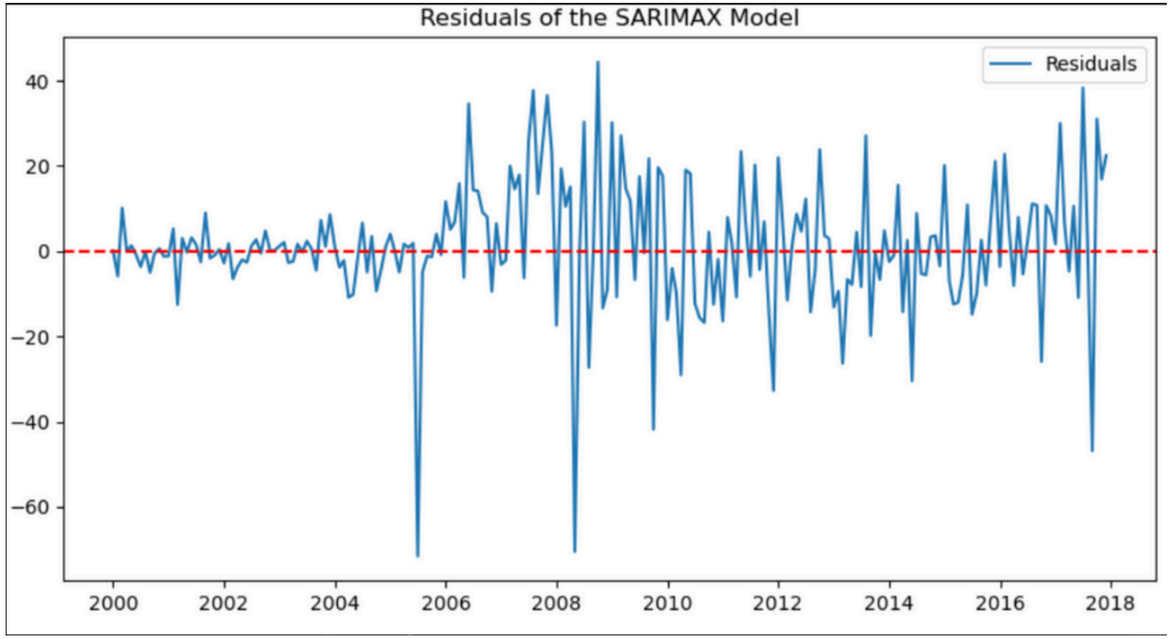
The Equation:

$$\cdot\ 0.0210L - 0.7114L^2)(1 - 0.5119L^{12})y_t = (1 - 0.1204L + 0.6832L^2)(1 - 0.3^2 - 0.3540L^{24})$$

After testing with various other SARIMA models, The best model is

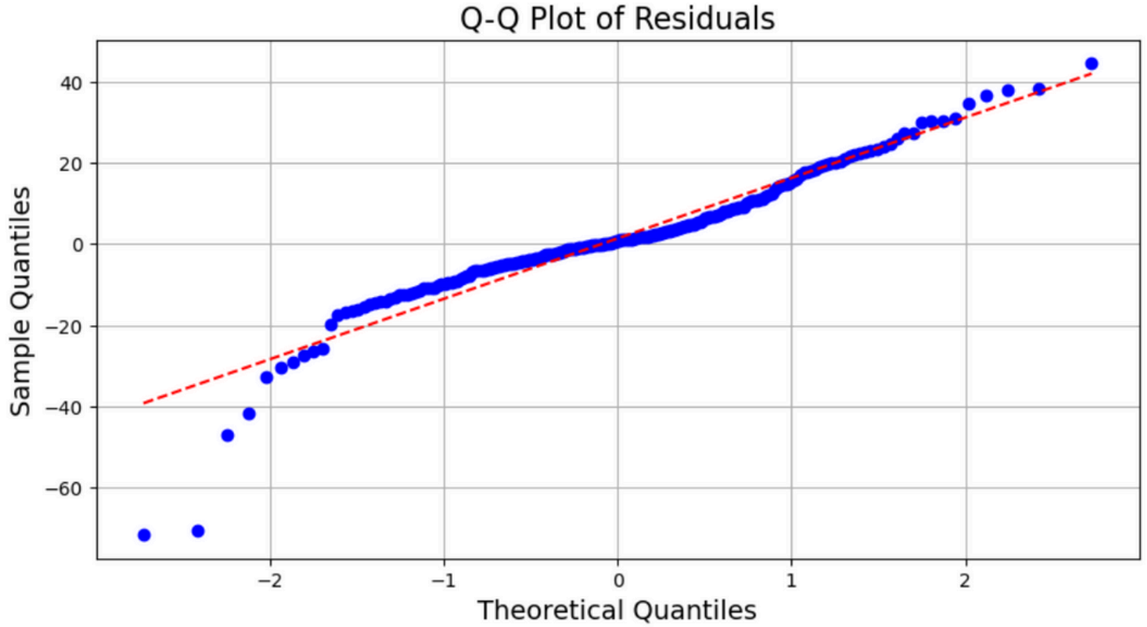
```
best model found: SARIMAX(2, 0, 2)x(1, 1, 2, 12) with AIC=1538.4865145609897, BIC=1576.6003113518
```

Model Adequacy

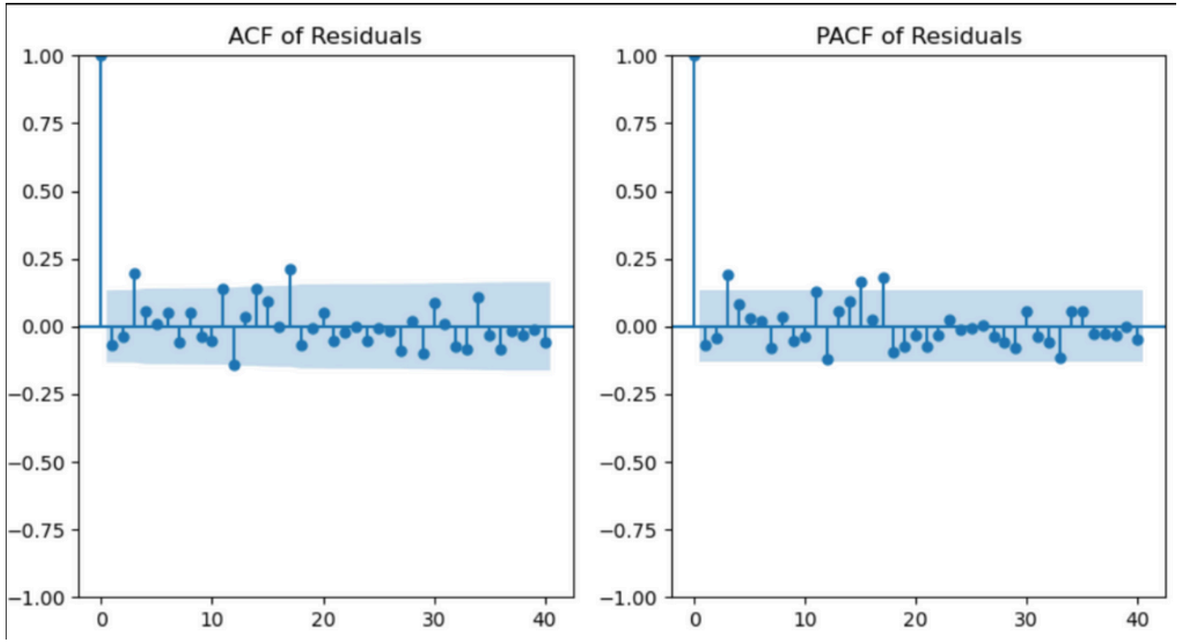


The residuals (errors) appear to fluctuate randomly around zero, which indicates that the SARIMAX model is capturing the trend and seasonality in the data reasonably well.

The Q-Q plot of residuals compares the quantiles of the residuals with the quantiles of a standard normal distribution.



Model Adequacy



The p-value is 0.1944, which is greater than 0.05. Since the p-value is greater than 0.05, we fail to reject the null hypothesis. This suggests that there is no significant autocorrelation in the residuals up to 10 lags.
This result implies that the residuals are likely independent over time, satisfying one of the key assumptions of the SARIMAX model.

Ljung-Box Test:

	lb_stat	lb_pvalue
10	13.552866	0.194381

